

# Introduction

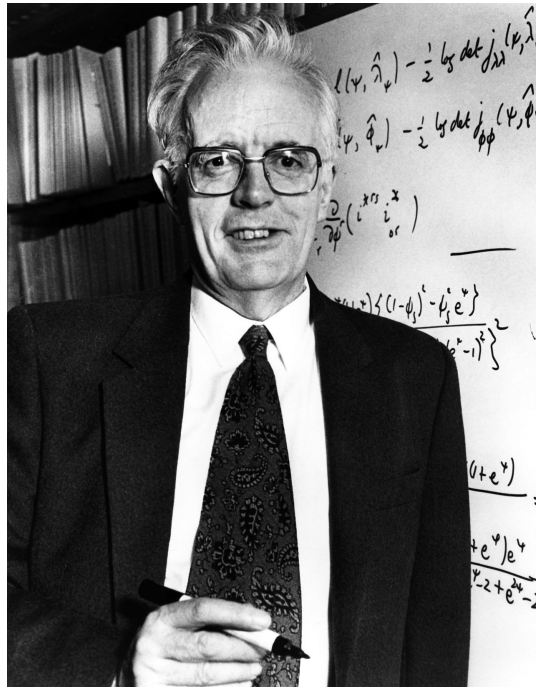
Statistics III - CdL SSE

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# Homepage



*"I would like to think of myself as a scientist, who happens largely to specialise in the use of statistics."*

Sir David Cox (1924-2022)

- *Statistica III* is a monographic course on **Generalized Linear Models** (GLMs), a broadly applicable **regression** technique.
- This is a **B.Sc.-level** course, but there are some prerequisites: it is assumed that you have already been exposed to:
  - **Simple linear regression**, from *Statistica I*;
  - **Inferential statistics**, from *Statistica II*;
  - **Linear models**, from *Analisi Statistica Multivariata* and *Econometria*;
  - **R software**, from *Analisi Statistica Multivariata*.
- In *Statistica III* we extend linear models within a unified and elegant framework.
- Regression is such an important topic that the **tour** will **continue** in the **M.Sc. at CLAMSES**. In **Data Mining** I will cover penalized methods and nonparametric regression.
- Indeed, GLMs can arguably be regarded as one of the most influential statistical ideas of the 20th century.

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# Statistics of the 20th century

*Biometrika* (2001), **88**, 1, pp. 13–52  
 © 2001 Biometrika Trust  
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## ***Biometrika* Centenary: Theory and general methodology**

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### SUMMARY

Contributions to statistical theory and general methodology published in *Biometrika*, 1901–2000, are telegraphically reviewed.

*Some key words:* Bayesian inference; Estimating function; Foundations of statistics; Generalised regression model; Graphical method; Graphical model; Laplace approximation; Likelihood; Missing data; Model selection; Multivariate statistics; Non-regular model; Quasilikelihood; Saddlepoint; Simulation; Spatial statistics.

- **Biometrika** is among the most prestigious journals in Statistics. Past editors include Karl Pearson, Sir David Cox, and Anthony Davison.

## Early ideas

- Classical linear models and least squares began with the work of **Gauss** and **Legendre** who applied the method to astronomical data.
- Their idea, in modern terms, was to **predict the mean** of a normal, or Gaussian, distribution as a function of a **covariate**:

$$\mathbb{E}(Y_i) = \beta_1 + \beta_2 x_i, \quad i = 1, \dots, n.$$

- As early as Fisher (1922), a more advanced non-linear model was introduced, designed to handle **proportion data** of the form  $S_i/m$ .
- Through some modeling and calculus, Fisher derived a **binomial** model for  $S_i$ , with

$$\mathbb{E}(S_i/m) = \pi_i = 1 - \exp\{-\exp(\beta_1 + \beta_2 x_i)\}, \quad i = 1, \dots, n.$$

where  $\pi_i \in (0, 1)$  is the **probability of success** of a binomial distribution.

- The corresponding inverse relationship is known as the **complementary log-log** link function:

$$\beta_1 + \beta_2 x_i = \log\{-\log(1 - \pi_i)\}.$$

## Early ideas II

- In the **probit model**, developed by Bliss (1935), a **Binomial** model for  $S_i$  is specified with

$$\mathbb{E}(S_i/m) = \pi_i = \Phi(\beta_1 + \beta_2 x_i), \quad i = 1, \dots, n.$$

where  $\Phi(x)$  is the cumulative distribution function of a standard Gaussian distribution.

- Dyke and Patterson (1952) also considered the case of modelling proportions, but specified

$$\mathbb{E}(S_i/m) = \pi_i = \frac{\exp(\beta_1 + \beta_2 x_i)}{1 + \exp(\beta_1 + \beta_2 x_i)}, \quad i = 1, \dots, n.$$

- The corresponding inverse relationship is known as the **logit** link function:

$$\beta_1 + \beta_2 x_i = \text{logit}(\pi_i) = \log \left( \frac{\pi_i}{1 - \pi_i} \right).$$

In fact, this approach is currently known as **logistic regression**.

- See Chapter 1 of McCullagh and Nelder (1989) for a more exhaustive **historical** account, including early ideas about the **Poisson** distribution, the multinomial, and more.

# Generalized linear models

## 10. GENERALISED REGRESSION

### 10.1. Generalised linear models

One of the most important developments of the 1970s and 1980s was the unification of regression provided by the notion of a generalised linear model (Nelder & Wedderburn, 1972; McCullagh & Nelder, 1989) and its associated software, though the concept had appeared earlier (Cox, 1968). In such models the response  $Y$  is taken to have an exponential family distribution, most often normal, gamma, Poisson or binomial, with its mean  $\mu$  related to a vector of regressor variables through a linear predictor  $\eta = x^T \beta$  and a link function  $g$ , where  $g(\mu) = \eta$ . The variance of  $Y$  depends on  $\mu$  through the variance function  $V(\mu)$ , giving  $\text{var}(Y) = \phi V(\mu)$ , where  $\phi$  is a dispersion parameter. Special cases are:

for fitting regression models. The estimating equations for a generalised linear model for independent responses  $Y_1, \dots, Y_n$  and corresponding covariate vectors  $x_1, \dots, x_n$  may be expressed as

$$\sum_{j=1}^n x_j \frac{\partial \mu_j}{\partial \eta_j} \frac{Y_j - \mu_j}{V(\mu_j)} = 0, \quad (10)$$

or in matrix form

$$D^T V^{-1} (Y - \mu) = 0, \quad (11)$$

where  $D$  is the  $n \times p$  matrix of derivatives  $\partial \mu_j / \partial \beta_r$ , and the  $n \times n$  covariance matrix  $V$  is diagonal if the responses are independent but not in general. Taylor expansion of (11)

- The **pivotal paper** by Nelder and Wedderburn (1972) unified all these approaches.

# Quasi likelihoods

## 10.2. Quasilikelihood

Data are often overdispersed relative to a textbook model. For example, although the variance of count data is often proportional to their mean, the constant of proportionality  $\phi$  may exceed the value anticipated under a Poisson model, so  $\text{var}(Y) = \phi\mu$  for  $\phi > 1$ . One way to deal with this is to model explicitly the source of overdispersion by the incorporation of random effects; see § 3.5. The resulting integrals can considerably complicate computation of the likelihood, however, and a simpler approach is through quasilikelihood (Wedderburn, 1974).

Quasilikelihood is perhaps best seen as an extension of generalised least squares. To see why, note that (11) is equivalent to  $U(\beta) = 0$ , where  $U(\beta) = \phi^{-1}DV^{-1}(Y - \mu)$ . Asymptotic properties of  $\hat{\beta}$  stem from the relations  $E(U) = 0$  and  $\text{cov}(U) = -E(\partial U / \partial \beta)$ , corresponding to standard results for a loglikelihood derivative. However, these properties do not depend on a particular probability model, requiring merely that  $E(Y) = \mu$  and  $\text{cov}(Y) = \phi V(\mu)$ , subject also to some regularity conditions. Hence  $\hat{\beta}$  has the key properties of a maximum likelihood estimator, namely consistency and asymptotic normality, despite not being based on a fully-specified probability model. Moreover, it may be computed simply by solving (11), that is, behaving as if the exponential family model with variance function  $V(\mu)$  were correct. The scale parameter  $\phi$  is estimated by  $\hat{\phi} = (n - p)^{-1}(Y - \hat{\mu})^T V(\hat{\mu})^{-1}(Y - \hat{\mu})$ , and the asymptotic covariance matrix of  $\hat{\beta}$  is  $\hat{\phi}(D^T V D)^{-1}$  evaluated at  $\hat{\beta}$ . A unified asymptotic treatment of such estimators from overdispersed

# The content of this course

- **General theory**
  - Linear models and misspecification
  - Generalized Linear Models (GLMs)
- **Notable models**
  - Binary and binomial regression
  - Poisson regression
- **Advanced topics**
  - Quasi likelihoods
- Unfortunately, due to time constraints, we will **not** cover:
  - Contingency tables and log-linear models;
  - Multinomial response and ordinal response models;
  - Models with correlated responses (random effects);
  - Nonparametric regression.
- These topics will be covered e.g. in **Statistica Multivariata** and **Data Mining** at CLAMSES.

# Textbooks

We will use several textbooks throughout this course — some more specialized than others. They are listed in order of importance:

1. The book by Salvan et al. (2020), **in Italian**, is the **main textbook**. Most of the material covered in these slides can be found there. I will also try to follow its notation as closely as possible.
2. The book by Azzalini (2008), **in Italian**, is more concise but very enjoyable to read. I highly recommend browsing through it.
3. The book by Agresti (2015), **in English**, is comprehensive and extremely well-written. It was the one I consulted most while preparing this course. Its only “drawback” is that it is in English.
4. The book by McCullagh and Nelder (1989), **in English**, is an **advanced and authoritative textbook** intended for experienced statisticians (at least at the M.Sc. level). Feel free to explore it out of curiosity, but it is not a main reference.

# Exam

- The written exam has two parts, held on the same day:
  - **Theory and exercises**: questions to assess understanding of concepts and the ability to correctly set up a statistical model.
  - **Data set analysis**: applied analysis of a dataset using R.
- The **overall mark** is the average of the two parts.
  - You must pass both parts (each  $\geq 18$ ).
- The **oral exam** is optional:
  - Can be requested by the student or the teacher
  - Final mark = average of written and oral marks
- The exam is **closed-book and closed-notes**, except for the **R scripts** provided at the beginning of the test.

# References

- Agresti, A. (2015), *Foundations of Linear and Generalized Linear Models*, Wiley.
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